Handling of Static Sensitive Electronic Components and Devices – General Data

1. Scope

This standard specifies the procedure and requirements for the protection of static sensitive electronic components and devices during receipt verification, transportation, storage, stores handling, assembly, assembly verification, installation and removal.

2. Equipment

A. Static awareness labels, self-adhesive, as per JEDEC–14 standard (e.g., 3M No. 7101), as shown in Figure 1.

B. Static shielding bags, transparent plastic, 6" X 10", capable of preventing static damage to static sensitive components or devices (e.g., 3M No. 2100–6–10).

C. Conductive foam, non-corrosive, high density, 1/4" thick (e.g. 3M No. 2910–1/4–12–12).

D. Static free cushioning material, plastic honeycomb cell construction to PPP–C–1842, Type III.

E. 3M #8020 static free work station grounding kit, components as listed below. Set up static free work stations as shown in Figure 2. As an alternative to the set-up shown in Figure 2, static free work stations may be set up as shown in Figure 3. It is acceptable for the wrist strap ground cord to be connected to the same ground lug as the table mat ground cord or directly to an electrical connection to earth ground. As an alternative to the 3M #8020 static free work station grounding kit, it is acceptable to equip static free work stations with static dissipative table mats, floor mats, wrist straps and grounding cords with a resistance 10^6 – 10^9 ohms; it is not acceptable to use conductive (10^5 or less ohms) or insulated (10^{10} or more ohms) table mats, floor mats, wrist straps or ground cords.

- 8210 Series Static Table Mat
- 8220 Series Static Floor Mat
- 2066 CHARGE-GUARD Wrist Strap (with ground cord)
- 3040 Ground Cord
- 3043 Inter-Connecting Ground Cord

F. Tote-Boxes, conductive plastic.

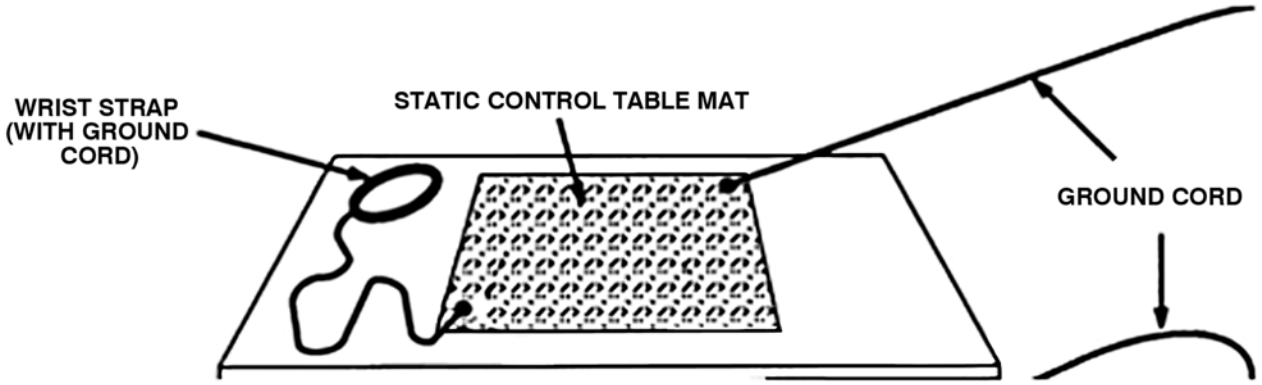
G. Connector caps, conductive plastic.

H. Resistivity meter (e.g., ACL model 375 or ACL model 475).

I. Wrist strap tester (e.g., Bystat Strapcheck BSL–9010).



Static Awareness Label
Figure 1



Set-Up of Static Free Work Stations
Figure 2



Alternative Set-Up of Static Free Work Stations
Figure 3

3. Procedure

A. General

1) Components or devices which can be damaged or destroyed by an electrostatic discharge (ESD), or shock, are termed static sensitive. Consider all components which are to be assembled to printed circuit boards (transistors, integrated circuit (IC) chips, etc.) static sensitive components. Consider all completed or partially completed circuit boards and avionics black boxes are static sensitive devices. Printed circuit boards on which no components have been assembled are not considered static sensitive.

2) A static electronic shock is caused by the instantaneous discharge of an electrostatic potential or voltage which is built through friction against a non-conducting surface, such as a nylon rug or a plastic chair. An ESD shock too small to even notice may be enough to destroy the electronic logic of an IC chip.

3) To prevent ESD damage to static sensitive components and devices it is necessary to protect from static shock at all times. During transport and storage, the static sensitive components and devices must be shielded from electrostatic fields by being packaged in ESD protective material. During receipt verification, stores handling, assembly and installation or removal of static sensitive components or devices, personnel must be grounded at all times by means of a conductive wrist strap. When installing or removing avionics boxes, personnel must be grounded to the aircraft structure by means of a wrist strap or other grounding equipment at all times.

B. Storage

1) Store static sensitive components in their original boxes, bags or static shielded tubes, identified by a static awareness label.

2) Place completed or partially completed printed circuit boards to be stored within a suitable static shielding bag identified and sealed by means if a static awareness label.

3) Ensure all connectors are capped with ESD protective caps and package avionics black boxes to be stored in static free cushioning material. Identify the package as static sensitive by means of a static awareness label.

C. Stores Handling

1) Unless it is necessary to split a lot or batch of components or devices, static sensitive components or devices may be issued from stores with no special handling procedures, except that the package must not be opened. If it is necessary to split a lot or batch, repackage the components or devices for issue as follows:

a) Place the original package of components or devices on a conductive table mat at a static free work station. Personnel filling the order must be grounded to the static free work station by means of a conductive wrist strap before opening the container in which the components or devices are packaged.

b) Repackage static sensitive components and devices as follows:

- Individually repackage static sensitive components, other than IC chips, in a suitable static shielding bag.
- Transfer IC chips onto conductive foam pads by placing the chip onto the foam pad and gently pushing down to press the chip leads partially into the foam and then place the foam pad in a static shielding bag. Take extreme care at all times when handling IC chips to avoid touching or bending the chip leads.
- Place printed circuit boards on which components have been assembled within a static shielding bag. Printed circuit boards on which no components have been assembled do not require ESD protective packaging.
- For avionics boxes, cap all connectors with ESD protective caps and package in static free cushioning material.

c) Seal static shielding bags or static shielding material and identify as static sensitive by means of a static awareness label placed in a readily visible location.

d) Place the identified static shielding bag or static cushioning material bag in a conductive plastic tote-box.

D. Installation and Removal of Avionics Boxes

1) Take care to avoid touching the connector contact pins when installing or removing avionics boxes.

2) Immediately after removing installed avionics boxes, fit conductive plastic caps over the connectors and package the avionics in static free cushioning material. Identify the static free cushioning material as containing a static sensitive component or device.